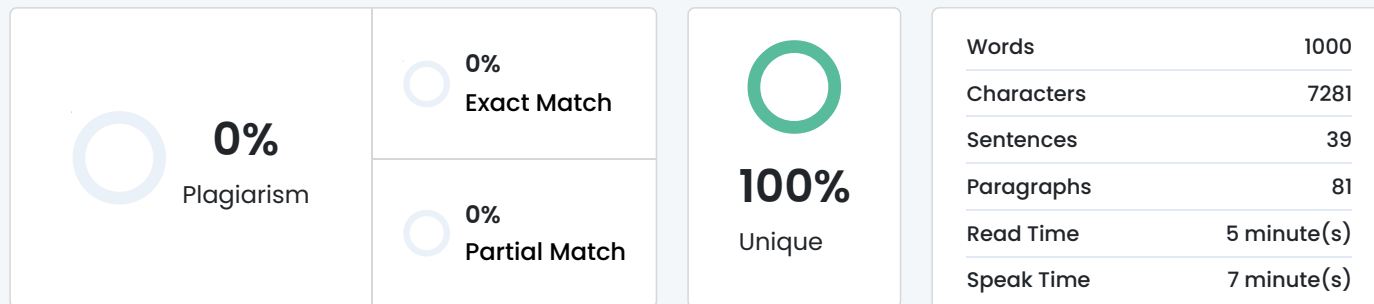


Plagiarism Scan Report



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Fuel Consumption Analysis System Using Python

REPORT

CSA0806– PYTHON PROGRAMMING

Course Outcomes: CO1,CO2, CO3 and CO4

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Academic Year: 2026–2027

CERTIFICATE

This is to certify that Rishiykesh K S (192511466) of CSA0806 has successfully prepared and submitted the report entitled "Fuel Consumption Analysis System Using Python" as part of the academic requirements for CSA0806-Python programming.

The work presented in this report addresses processor-based Fuel Consumption, peripheral interfacing, signal control, interrupt handling, timing, emergency-vehicle priority, data storage and serial communication as specified in the assignment.

Faculty Signature: _____

Date: _____

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Abstract

The Fuel Consumption Analysis System is a menu-driven Python application designed to help individuals and small fleet operators monitor and analyse vehicle fuel usage efficiently. The system allows users to register vehicles, record trip and refuelling details, and calculate important fuel-efficiency metrics such as mileage, total fuel consumption, monthly consumption, and fuel cost. All records are stored persistently using CSV files, ensuring that data remains available even after the program is restarted.

The system uses Python data structures such as lists, dictionaries, tuples, and sets for efficient data management and validation. It also separates data storage, analysis, and user interaction into independent modules, making the application easier to maintain and test. The system generates consolidated reports that rank vehicles according to fuel efficiency and identifies trips with abnormally high fuel consumption using a vehicle-specific threshold. Testing confirmed that the calculated mileage, summaries, rankings, and anomaly detection results matched independently calculated values. The project demonstrates the practical application of Python programming concepts to solve a real-world fuel monitoring and sustainability problem.

Introduction

Fuel consumption is an important factor in vehicle operation because it directly affects operating costs, resource usage, and environmental impact. For individuals and small fleet operators, maintaining accurate records of fuel usage can help identify inefficient vehicles, wasteful trips, and changes in vehicle performance. However, when fuel records are maintained manually in notebooks or spreadsheets, the information can become scattered, inconsistent, and difficult to analyse. The Fuel Consumption Analysis System addresses this problem by providing a simple, menu-driven console application developed using Python. The system enables users to register vehicles and record trip and refuelling information such as distance travelled, fuel filled, odometer reading, date, and fuel price. Using these records, the application calculates vehicle mileage, total fuel consumption, monthly consumption, and estimated fuel costs.

The system also provides a consolidated fleet report that ranks vehicles from most to least fuel-efficient and identifies trips with unusually high fuel consumption. To achieve this, the application uses separate modules for data persistence and analysis, while CSV files are used to store records between program executions.

Python concepts including control flow, string handling, functions, lists, dictionaries, tuples, sets, file handling, and modules are applied throughout the system. The project demonstrates how fundamental Python programming concepts can be combined to create a practical data-analysis application. By providing clear and measurable information about fuel usage, the system can help users make better decisions regarding vehicle maintenance, driving practices, and fuel efficiency while also supporting broader sustainability objectives.

Problem Statement and Problem Formulation

Manual record-keeping of vehicle fuel usage in notebooks or spreadsheets makes it difficult for individuals and small fleet operators to identify inefficient vehicles or wasteful trips in time to act on them. Readings are scattered, units are inconsistent, and no consolidated view exists to compare vehicles against each other or against their own historical performance.

This assignment formulates the problem as the design and implementation of a menu-driven Python console application that allows a user to: (a) register vehicles, (b) log refuelling and trip entries (distance travelled, fuel filled, odometer reading), (c) derive metrics such as mileage (km/l), monthly consumption and estimated cost from current fuel prices, and (d) generate a consolidated report that ranks vehicles by efficiency and flags trips with abnormally high consumption. The problem is decomposed into three sub-problems: reliable persistent storage of records, correct derivation of efficiency metrics from raw trip data, and a usable, error-tolerant command-line interface that ties the two together.

Objective and Expected Outcomes

- * Build a menu-driven console application that supports vehicle registration, trip/refuelling logging, per-vehicle summaries and a consolidated fleet report.
- * Apply control-flow constructs, string-handling routines and well-scoped functions to validate every user input before it is stored (CO2).
- * Model vehicles, trips and derived summaries using lists, dictionaries, tuples and sets, choosing each structure for the operation it performs best (CO3).
- * Persist all records to disk using file handling, and separate data-access code from analysis code into

independent, importable modules (CO4).

Expected outcome: a working application that, given a small set of registered vehicles and logged trips, correctly computes each vehicle's overall mileage and monthly fuel consumption, ranks vehicles from most to least efficient, and automatically flags trips whose fuel usage is significantly above that vehicle's own average – all confirmed against hand-calculated values in Section 9 (Test Cases).

Requirements, Constraints and Assumptions

Functional Requirements

- * Register a vehicle with a unique registration number, make, model and starting odometer reading.
- * Log a trip/refuelling event against a registered vehicle with date, distance, fuel filled, odometer and fuel price.
- * View a per-vehicle summary: trip count, total distance, total fuel, overall mileage, total cost and month-wise consumption.
- * Generate a consolidated report ranking all vehicles by efficiency and listing trips with abnormally high consumption.
- * Persist all data between runs using CSV files.

Constraints

- * Console-only interface (no GUI) built using core Python and the standard library only – no third-party packages.
- *

Matched Source

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